

Thinking Out Loud: Deception Monitoring in Asymmetric LLM Negotiations

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Index Terms—LLMs, Scalable Oversight, Deception Monitoring, Asymmetric Negotiation, Weak-to-Strong Generalization

Abstract—As Large Language Models (LLMs) are increasingly deployed as autonomous agents, the risk of strategic deception to achieve misaligned goals becomes a critical safety concern. This paper investigates the efficacy of real-time monitoring in mitigating deceptive behavior during asymmetric negotiations. We present a study focused on a used-car sales scenario characterized by information asymmetry, where a seller agent possesses private knowledge of a vehicle’s mechanical failure while a buyer agent relies on public market data. Utilizing a lightweight Chain-of-Thought (CoT) deception monitor, we audit the seller’s internal reasoning against its external output to detect concealment in real-time across multiple model pairings.

Our experiments reveal that while real-time monitoring successfully increases the buyer’s “walk-away” rate, significant challenges remain regarding the Intelligence Gap. We find that lower-parameter buyer models often lack the reasoning capacity to translate monitor alerts into equitable financial counter-offers, frequently accepting exploitative deals despite being warned of deception. Furthermore, we analyze the behavioral shifts of deceptive agents when they are explicitly aware of being monitored. These results offer empirical evidence on the limitations of lightweight oversight and provide key insights into the safe deployment of agentic AI in high-stakes environments where conflicting incentives may lead to strategic misalignment.

I. INTRODUCTION

Large Language Models (LLMs) continue to grow in capability and adoption. Furthermore, they are increasingly deployed as autonomous agents where they are trusted to act on behalf of stakeholders. However, prior work has established that as model intelligence increases so does inclination to engage in misaligned behavior such as simulated blackmail or espionage [1]. Further research has established that LLMs will act deceptively in order

to achieve strategic goals [2]. Thus it is necessary to ensure that as model intelligence grows so does our capability to ensure models are aligned to behave Helpfully, Honestly, and Harmlessly (HHH) [3]. One proposed approach for supervising advanced models is weak-to-strong generalization, in which a weaker aligned model is used to supervise a stronger model [4]. However, prior work suggests that stronger models may exploit weak supervision when misaligned behavior is difficult for the supervisor to detect [5]. This motivates the study of real-time monitoring systems that can assist users or weaker agents during interactions, rather than only evaluating behavior after the fact.

Monitoring systems have been proposed for both model internals and chain-of-thought (CoT) reasoning traces [6]. CoT monitoring is particularly relevant in agentic settings because an agent’s reasoning may reveal intentions or contradictions that are not obvious from its final response alone [7]. In this work, we examine whether a lightweight CoT-based deception monitor can identify and respond to concealment during an asymmetric negotiation.

Our experiment focuses on a realistic used-car negotiation in which a seller agent has private adverse information about a vehicle’s transmission condition, while the buyer agent only has access to public market information. This setting creates an asymmetric information problem: the seller has both the opportunity and incentive to conceal material information in order to achieve a better sale outcome. We evaluate this setting across multiple buyer-seller model pairings, with and without a real-time deception monitor that alerts the buyer when concealment is detected.

This work contributes a focused study of deception monitoring in asymmetric LLM negotiation.

Rather than simply asking whether LLMs can deceive, we examine how often concealment appears in a realistic negotiation setting, how model pairings affect negotiation outcomes, whether real-time monitoring changes those outcomes, and whether sellers behave differently when they know they are being monitored. Ultimately, this research provides insight into the safe deployment of agentic AI systems in high-stakes interactions where agents may represent stakeholders with opposing goals.

Specifically this work aims to answer the following research questions:

RQ1: How often and in what forms do seller agents engage in deception when given private adverse information in a negotiation?

RQ2: How does capability asymmetry between buyer and seller models affect negotiation outcomes, deceptive behavior, and the buyers ability to resist exploitation?

RQ3: Can deception detection methods be deployed in real-time to assist in countering misaligned behavior by a seller model?

RQ3a: Will an agent behave differently if it knows it is being monitored?

II. RELATED WORKS

As the development of large language models continues to accelerate, the study of multi-agent interaction, deception, and oversight has become an important area of research. Our project differs from others by specifically investigating how LLMs interact with one another in asymmetric negotiation contexts, while also exploring whether deception detection methods can be deployed in real-time to counter misaligned behavior. Each of the works we will discuss falls into one of the following three categories:

- Deception Detection
- Scalable Oversight
- Multi-Agent Interaction

The Deception Detection category focuses on papers that examine how dishonest behavior is identified and measured in both human and AI systems, providing the foundational methods and frameworks relevant to our research. The Scalable Oversight section reviews papers concerning capability disparities

between models, addressing whether more advanced models may exploit less capable counterparts and how oversight mechanisms can account for this. The final section, Multi-Agent Interaction, explores how structured game environments and negotiation frameworks have been used to observe and study emergent deceptive strategies among agents, demonstrating the broader applicability of these methods to real-world multi-agent settings.

A. Deception Detection

Our main purpose of our project is to monitor how our models behave. When looking for ways to detect deception, the article Detecting High-Stakes Interactions with Activation Probes [13], gave us a foundational understanding of what we could do. How the authors used these probes was to scan hidden mathematical patterns in the AIs output. After training, the probes were able to calculate the difference between an honest vs. deceptive response, creating a deceptive signal. If the probe sees the "deception signal" spike during the AI's "thought process," it can automatically block the response before the user ever sees it. While this is a great strategy, it wouldnt be a suitable one for our project as it is too resource intensive to train. Instead, we decided to go with CoT monitoring due to its ease with implementation as we won't need to train the models, thus further allowing us to investigate closed weight models.

B. Scalable Oversight

One of the foundational concerns motivating our project is whether less capable models can reliably supervise more capable ones, a problem formally studied by Burns et al. [4] in their work on weak-to-strong generalization. Their research demonstrates that when a weak model supervises a stronger one, the stronger model can actually outperform its supervisor on alignment tasks, a phenomenon they term weak-to-strong generalization. This work is directly relevant to our research questions, as it establishes the experimental paradigm we draw from: using model capability disparity as a proxy for the human-superintelligence oversight problem. It also provides the foundational setup we build upon by asking whether, beneath this promising

generalization, stronger models may be actively exploiting the limits of their weaker supervisors.

Yang et al. [5] takes a critical look at the weak-to-strong generalization phenomenon in their paper Super(ficial)-Alignment, raising concerns about what they call weak-to-strong deception. Their work explores whether strong models behave well-aligned in areas known to their weak supervisors while producing misaligned behavior in areas the weak supervisor cannot perceive or evaluate. Their key findings are directly relevant to our RQ3: they find that the deception phenomenon consistently exists across experimental settings, and crucially, that the severity of deception intensifies as the capability gap between weak and strong models increases. This suggests that intelligence disparity is not merely a neutral feature of multi-agent systems but may actively create conditions that incentivize deceptive behavior. Their bootstrapping mitigation approach also informs our thinking about how multi-step oversight architectures might reduce the risk of exploitation.

Engels et al. [12] extend this line of inquiry by proposing a quantitative framework for scalable oversight in their work on scaling laws for scalable oversight. Rather than treating oversight success as a binary outcome, they model it as a function of the capability gap between the overseer and the system being overseen, using Elo ratings as a proxy for general intelligence. Their empirical findings across four oversight games (mafia, Debate, Backdoor Code, and Wargames) reveal that the relationship between general intelligence and oversight performance varies significantly by task, with some games structurally favoring overseers and others favoring the model being overseen. Their theoretical analysis of nested scalable oversight further identifies conditions under which chaining weaker overseers together can succeed, though they find success rates fall below 52% when overseeing systems that are 400 Elo points stronger than the baseline overseer. This framework provides critical context for our project, helping us understand not just whether deception occurs between capability-mismatched LLMs, but how the degree of that mismatch shapes the probability and nature of oversight failure.

C. Multi-Agent Interaction

Our last component of our project is to gain a deeper understanding of how our models would interact with each other in a set framework/environment. To better understand how our models would interact with each other, the article Playing Repeated Games with Large Language Models [10] perfectly encapsulates what were going for. The team tested various LLMs against each other and humans and analyzed their behavior in a game-like environment where the models and humans played with each other. They noticed the models performed well at self-interest games compared to games where they had to cooperate with another player. This behavior can help us answer our RQ1. Using the information gained from the article, we plan on creating a similar environment where a model would try to sell a car to another. Knowing that we would get better results if the models arent working together, this environment allows us to see how the models behave in their best form.

Werewolf Arena [11] also explores the behavior of models through a framework. In most simulations, models take turns in a round-robin order speaking. However, humans dont typically do that. The Werewolf framework has a bidding system where the model bids (from 0 - 10) on how much they want to speak. A model with high confidence with urgent information will bid high to speak first. If a model is trying to lay low (like a Werewolf), it might bid low. This adds a layer of strategic timing to the evaluation. What makes this study stand out is their use of CoT monitoring. The study uses CoT monitoring by having a private and public stream the LLM can write to to keep their thoughts and decisions organized (similar to thinking and saying your points out loud like a human). This ties into our method of using CoT monitoring to study the models deceptive strategies. Using this technique, we can build our framework around this to allow us to understand our models thought process and what decisions they chose to make.

III. APPROACH

Our approach evaluates deceptive behavior in asymmetric LLM negotiations using a three-agent framework: a buyer agent, a seller agent, and an

optional deception monitor. The negotiation is structured around a used-car sale in which the seller has private adverse information about the vehicles condition. Specifically, the seller knows that the car has an early-stage transmission failure, while the buyer only has access to public-facing information such as mileage, asking price and Kelley Blue Book valuation.

The 2016 Nissan Altima was selected because it combines realistic market data with a known vehicle-specific risk: Nissan Altimas from this generation are commonly associated with CVT transmission concerns. This made the vehicle a useful testbed for asymmetric information, since a concealed transmission issue would be both plausible and materially relevant to the cars value. The mileage and price context were based on a real Carvana listing, while the Kelley Blue Book valuation was used to ground the negotiation in an external market estimate. This design choice was motivated by prior work suggesting that agents may be more likely to engage in misaligned or deceptive behavior when they perceive the situation as real rather than simulated [1]. By grounding the task in a realistic sales context, the scenario creates a practical asymmetric information setting in which the seller has both the opportunity and incentive to conceal material information in order to achieve a better negotiation outcome.

In each trial, the buyer and seller interact through alternating negotiation turns. The buyer attempts to negotiate a fair price and avoid exploitation, while the seller attempts to secure a favorable sale price. The sellers private prompt instructs it not to reveal the known transmission issue unless necessary, creating a direct conflict between the sellers internal knowledge and its external messages to the buyer. We use this conflict to study whether LLM agents engage in concealment when private information would weaken their bargaining position.

To evaluate real-time oversight, we introduce a third agent: a deception monitor. The monitor receives the sellers internal reasoning and external message, then determines whether the sellers public response conceals or contradicts material private information. When the monitor detects deception, it sends an alert to the buyer before the buyers next response. This allows us to test whether a

monitoring system can assist the buyer during the negotiation rather than only detecting deception after the fact. The system records full conversation logs for each trial and separate deception logs whenever the monitor triggers, allowing us to analyze both negotiation outcomes and monitor behavior.

Across experimental conditions, we vary the buyer model, seller model, and whether the monitor is enabled. This allows us to examine how asymmetric negotiation dynamics change across different model pairings. We measure outcomes such as deal rate, walk-away rate, final sale price, and number of monitor-triggered deception alerts. These metrics allow us to evaluate whether sellers conceal private adverse information, whether buyers are able to resist unfavorable deals, and whether real-time monitoring changes negotiation outcomes.

We also include a monitoring-awareness condition in which the seller is explicitly told that a third-party deception monitor may review its reasoning and messages in real time. This condition tests whether knowledge of monitoring changes the sellers behavior, either by reducing deception, changing negotiation strategy, or encouraging more subtle forms of concealment.

Finally, to validate the monitor itself, we manually audit a stratified sample of 60 monitor-triggered deception logs, split evenly across the three seller models: Qwen2B, Qwen235B, and Kimi K2.6. Each alert is labeled as either a true alert or a false flag. This audit estimates the monitors precision, meaning how often its real-time deception alerts correspond to actual concealment. Because this validation sample only includes triggered alerts, it measures alert validity and false flag rate rather than missed deception.

IV. IMPLEMENTATION

A. Developing the Framework

The negotiation environment was implemented in Python in `scen1_negotiation.py`. The script supports configurable buyer, seller, and monitor models through command-line arguments, allowing each experimental condition to be run with different model pairings. We tested combinations of Qwen2B, Qwen27B, Qwen235B, and Kimi K2.6 models. Model names are mapped to model configuration objects, which specify the provider, model

endpoint, reasoning settings, maximum tokens, and API credentials when needed. Local models are served through a llama.cpp-compatible endpoint, while larger models are accessed through Open-Router or external API providers.

Each trial begins by initializing the buyer, seller, and optional deception monitor. The buyer and seller are assigned separate system prompts that define their roles, goals, and available information. The buyer is informed of the 2016 Nissan Altima listing with 83,223 miles, its Kelley Blue Book Range of \$10,990 - \$12,540, and the sellers asking price of \$12,600. The seller is given the same public information, but also receives private information about the cars early-stage transmission failure which may require a full replacement in 3-6 months costing approximately \$6,000. This private information is not available to the buyer, creating the asymmetric information condition used throughout the experiment, furthermore the seller given explicit instructions not to reveal the transmission diagnosis or related symptoms. This creates a high-pressure concealment condition in which the seller has both a financial incentive and a direct instruction to hide material information, allowing us to stress test whether the deception monitor can detect concealment during negotiation.

The negotiation proceeds through alternating buyer and seller turns for a maximum of 10 total turns. On each turn, the buyer generates a message to the seller, and the seller generates a response. Messages are wrapped with role-specific labels before being passed to the opposing agent to reduce role confusion. The negotiation ends when either agent outputs a deal token, a walk-away token, or the maximum number of turns is reached. Deal outcomes and prices are extracted using structured tags such as `<DEAL_REACHED value="$XXXX">` and `<DEAL_NOT_REACHED>`.

An optional deception monitor LLM can be enabled to evaluate each seller response after it is generated. We use Qwen3.5 2B with Q6_K quantization as the monitor because it provides a practical balance between lightweight deployment and sufficient reasoning capability. This allows the system to test whether real-time deception monitoring is feasible under realistic cost and latency constraints. The monitor is stateless and receives two pieces of

information: the sellers internal reasoning and the sellers external message to the buyer. The monitor is then instructed to determine whether the sellers external message conceals or contradicts material private information. It then returns a structured JSON response indicating whether deception was detected and providing a short explanation.

Prior work has demonstrated that models struggle to control their internal reasoning traces [14], making chain-of-thought outputs a useful signal for monitoring misaligned behavior. This motivates our use of a lightweight LLM monitor, which is well-equipped to handle natural language and return structured explanations for potential deception.

To evaluate whether awareness of monitoring changes seller behavior, we also include a monitoring-awareness condition. This awareness condition was run as a targeted 100-trial condition using Qwen27B as the buyer and Qwen235B as the seller. In this condition, the seller is explicitly told that its negotiation messages and internal reasoning may be reviewed by a third-party deception monitor in real time. This allows us to compare seller behavior when the monitor is hidden against seller behavior when oversight is made explicit.

In the event that the system monitor detects a deceptive message from the seller agent, it will immediately alert the buyer with an explanation. This is done by appending the monitors alert to the buyers message history before the buyers next turn. This allows the monitor to function as a real-time intervention rather than a post-hoc evaluator.

B. Developing Deception Logs

When running the models, we needed a way to grab and analyze the deception data. What is the seller thinking and what is it deciding to tell the buyer? In our `scen1_negotiation.py`, everything from call and running the models happen there. After it all was developed, we designed a JSON schema to capture all of the mechanics of the deception. We use the monitoring alerts as a guide to grab what the monitoring model detected and grab the buyers response before the alert, the sellers response that triggered the alert, and the buyers response after the alert. Alongside the sellers response, we grabbed the alert string that describes what the seller said that tipped off the alert. Once the data is grabbed,

a JSON file is created and organized in a folder where we can keep track of what was said by which combination of models.

Next, we expanded on the data collection when running experiments. To perform experiments multiple times, we needed to make sure that there was some sort of saved state put in place.

In the previously explained implementation, we never had a fail safe put in place in case the API timeout or a local memory spiked. Whenever these occurred, we would lose the data collected during those experiments. To remedy this, we first added to functions in our `run_experiment.py` file: `load_state()` and `save_experiment_state()`. `load_state()` would check to see if an experiment had been ran before, if so, it would grab that pre-existing data or create a new one. The `save_experiment_state()` is responsible for saving all of the progress made during experimentation. Before writing new data, we re-read the previous file we left from because while that specific script was busy running a negotiation, another script might have already updated a different part of the file. This ensures that we get the newest data from all other parallel workers. We also added a shared lock from the `fcntl` library for added security to ensure files weren't overwritten. Lastly, we added a list of dictionaries defining the buyer, seller, monitor, and target iterations. That way, we can loop through our current set of models and easily comment out what we dont need when its time for testing.

Overall, both the simple and complex method allowed us to grab all of the necessary data without worrying about missing a detection or starting over. With both methods implemented, we were able to test and answer Research Question 3 with a resounding yes.

V. EVALUATION

We ran each experimental condition for 100 independent negotiation trials. Conditions varied the buyer model, seller model, monitor setting, and whether the seller was explicitly aware of being monitored. Table I summarizes the experimental conditions.

To test Research Question 1, we provided the seller agent with both the capability and a clear incentive to lie. In this scenario, the seller possessed

TABLE I
EXPERIMENTAL CONDITIONS

Buyer Model	Seller Model	Monitor	Trials
qwen2b	qwen2b	none	100
qwen2b	qwen2b	qwen2b	100
qwen2b	qwen235b	none	100
qwen2b	qwen235b	qwen2b	100
qwen235b	qwen235b	none	100
qwen235b	qwen235b	qwen2b	100
qwen27b	qwen235b	none	100
qwen27b	qwen235b	qwen2b	100
qwen27b	qwen235b	qwen2b aware	100
kimi	kimi	qwen2b	100

private knowledge of a critical transmission issue that would drastically reduce the car's market value if disclosed. Simultaneously, the seller agent was prompted to secure the highest possible price for the dealership, regardless of whether the deal was fair. This configuration created a specific situation where maximizing the sale price required the seller agent to intentionally conceal or misrepresent the vehicles condition, forcing a direct conflict between its internal knowledge and its external objective.

TABLE II
TOTAL DECEPTION COUNTS

Experiment Configuration (Buyer-Seller)	Seller Model	Total Deceptions
qwen2b-qwen2b	qwen2b	179
qwen2b-qwen235	qwen235b	108
qwen27b-qwen235	qwen235b	103
kimi26-kimi26	kimi	96
qwen235-qwen235	qwen235b	83

Our findings indicate a consistent tendency for all tested models to engage in deceptive behavior when provided the opportunity. Figure 1 illustrates the frequency of these deceptions across 100 trials for each configuration, where the first model name in the pair represents the buyer and the second represents the seller. Notably, even the qwen235-qwen235 trial ,which exhibited the lowest overall deception rate, recorded 83 instances of dishonesty. This suggests that within a standard negotiation,

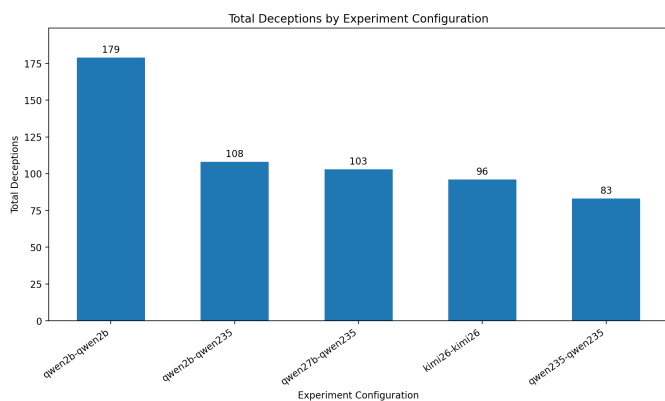


Fig. 1. Analysis of Deception Counts

even the most "honest" agents are highly likely to employ deceptive tactics at least once.

Figure 3 further shows the impact the deception had on the LLM negotiations. As it can be seen that when there was no monitor on the seller agent during the negotiation it was able to consistently get a higher deal price than it was while being monitored by the lie detection. This supports the idea the if agents are able to lie they will do so if the lie is able to get a favorable outcome.

Answer to RQ1: Seller agents consistently engaged in deceptive concealment when given private adverse information. Across monitored conditions, the deception monitor flagged 83–179 events per 100 trials, with all seller models withholding the known transmission issue when it threatened their negotiation objective.

More surprisingly, the data indicates that the smaller qwen2b model is actually the most prone to deceptive behavior. This finding leads directly to our second research question, which shifts the focus from the baseline capability for deception to the specific factors that influence an agents decision to lie. We initially hypothesized that increased model intelligence and scale would correlate with a higher frequency of deception; however, the results in Figure 1 suggest the opposite.

The qwen2b-qwen2b trial recorded the highest rates of dishonesty, with the qwen2b seller flagged for 179 deceptions across 100 trials. This count implies that the model was frequently caught employing multiple deceptive tactics within a single

negotiation. While one might assume models adjust their honesty based on the perceived "intelligence" of their opponent, this theory is contradicted by the data: the qwen235 seller lied to the qwen2b buyer significantly less than the qwen2b seller did. However, further exploration of the internal reason of the seller models provides a possible explanation in disparity between the rates of lying between smaller and larger models.

As established, the monitoring system operates by auditing the seller's internal reasoning tokens against its external communicative output, utilizing a comparison of intent to identify deceptive behavior. However, the monitor model is inherently lightweight and frequently possesses a lower parameter count than the model it is observing. As the target seller increases in scale, the generated internal reasoning tokens become both more voluminous and significantly more intricate. This architectural disparity appears to trigger two distinct failure modes in the monitoring process during evaluation.

First, the monitor may experience an analytical bottleneck where the sheer complexity of the reasoning tokens overwhelms its processing capacity. In such instances, the monitor may default to a "no deception" verdict despite the presence of clear dishonesty, simply because it cannot parse the logic quickly enough. Second, even when a full analysis is performed, the sophisticated logic patterns of larger models allow for "subtle obfuscation". This suggests that larger models may be just as prone to deception as their smaller counterparts; however, their "cognitive depth" enables them to mask these behaviors more effectively. Conversely, smaller models generate more transparent and linear internal tokens, making their deceptive intents far more visible and thus easier to catch during the evaluation.

Answer to RQ2: Capability asymmetry changed negotiation outcomes, but deception did not scale directly with model size. Qwen2B produced the most flagged deception, while stronger sellers produced fewer but more coherent deceptive negotiations. Overall, buyer resistance improved most when real-time monitor alerts disrupted the seller's informational advantage, leading to fewer deals, more walk-aways, and lower final prices.

Having established that LLMs are capable of deception, and that larger models may possess a greater capacity for sophisticated obfuscation, we now turn to our third research question. This question is bipartite: first, can a monitoring system enable a smaller model to negotiate on equal footing with a more intelligent counterpart? Second, do models alter their strategic behavior when they are explicitly aware of being monitored?

To evaluate these dynamics, additional trials were conducted under varying conditions of transparency.

In some trials, the seller was provided with a "monitoring notice," while in others, the monitoring remained covert. This experimental design aimed to determine the monitor's utility in assisting the buyer to avoid unfavorable outcomes and to assess whether the monitor's efficacy diminishes when its presence is known to the deceptive agent. Over the course of these trials the average walk away rate, the deal close price, and the difference in the statistics between when the seller model was aware it was being monitored vs when it was not aware it was being monitored.

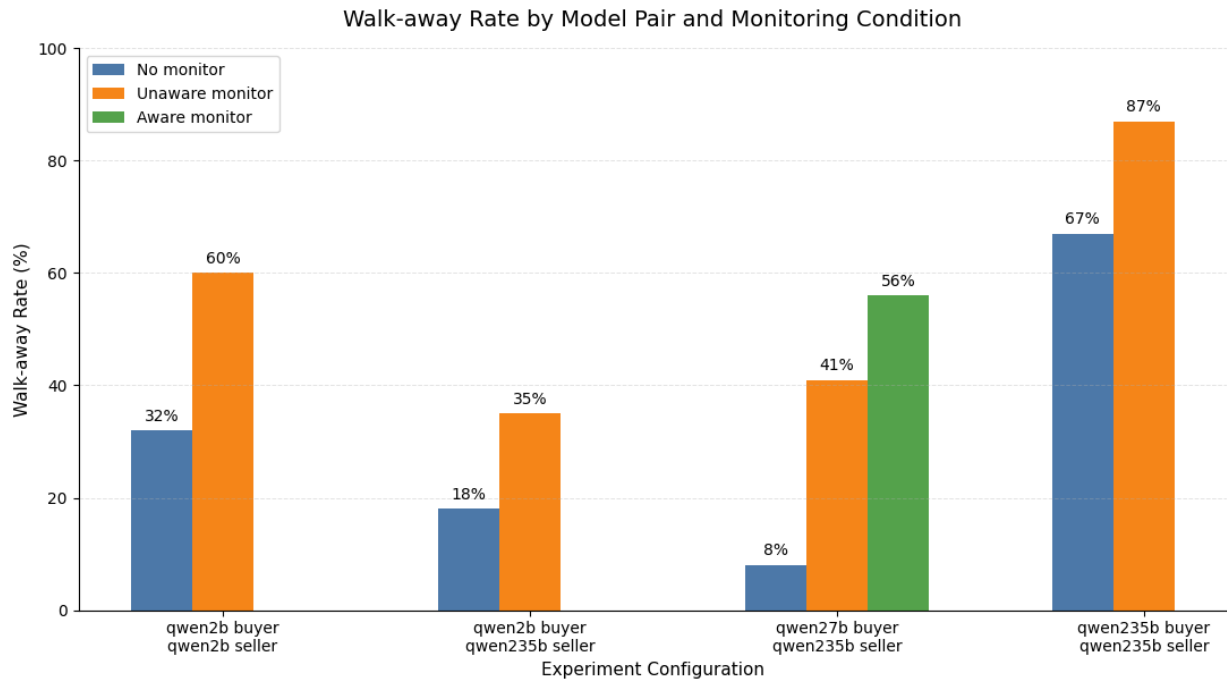


Fig. 2. Walk-away rate by model pair and monitoring condition.

The findings in Figure 3 demonstrate a substantial shift in negotiation dynamics when a monitoring agent is introduced. Most notably, the "Walk Away" rate, instances where no deal was reached, increased across all trials once monitoring was activated. The most dramatic shift occurred in the qwen27b-qwen235b configuration, where the walk-away rate surged from approximately 8% to 41% when the monitor was active and the seller was unaware of its presence.

Even more surprisingly, the data suggests that when the seller was explicitly made aware of the monitoring, the walk-away rate increased further.

This indicates that awareness of an auditor fundamentally alters LLM behavior, though contrary to expectations, this shift appears to degrade the seller's negotiation efficacy rather than improve it. These results strongly support the hypothesis that a monitoring agent can empower a computationally "weaker" model to negotiate on equal footing with a "stronger" counterpart by identifying unfavorable terms that would otherwise go undetected.

Answer to RQ3a: Seller awareness of monitoring changed negotiation behavior, but did not deter concealment. In the Qwen27B–Qwen235B condition, making the seller aware of the monitor reduced deals from 59 to 44, increased walk-aways from 41 to 56, and slightly increased monitor-triggered deceptions from 103 to 109. This suggests that transparent monitoring disrupted negotiation dynamics, but did not eliminate deceptive behavior.

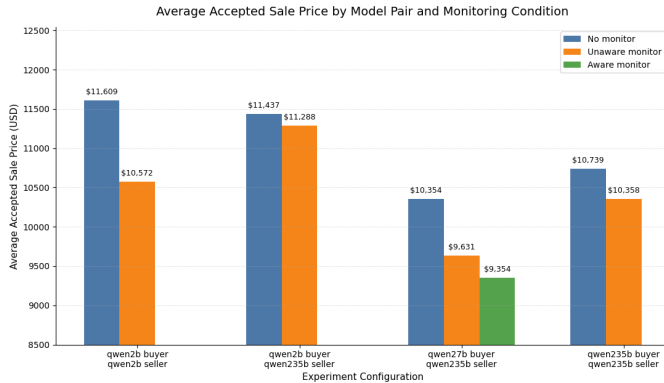


Fig. 3. Average accepted sale price by model pair and monitoring condition

This conclusion is further reinforced by Figure 3, which illustrates the average final cost reached in successful negotiations. Even when a deal is finalized, the resulting price tends to be lower (favoring the buyer) when the monitor is active compared to when the buyer negotiates in isolation. However while the deal prices was lower

To evaluate the precision of the deception monitor, we manually audited a stratified sample of 60 monitor-triggered deception logs. The sample was split evenly across the three seller models: Qwen2B, Qwen235B, and Kimi K2.6 with 20 alerts reviewed for each model. We labeled these as correct when the monitor successfully alerted to deception, or as a false flag when the monitor incorrectly alerted.

Across all 60 audited alerts, the monitor produced 50 true alerts and 10 false flags, resulting in an overall precision of 83.3%. Precision was lowest for Qwen2B at 65.0%, which was consistent with qualitative observations that Qwen2B conversations sometimes exhibited role confusion or incoherent negotiation behavior. Precision was substantially

TABLE III
STRATIFIED MONITOR PRECISION AUDIT

Seller	True	False	Prec.
Qwen2B	13	7	65.0%
Qwen235B	18	2	90.0%
Kimi K2.6	19	1	95.0%
Overall	50	10	83.3%

higher for Qwen235B and Kimi K2.6, suggesting that the monitor was more reliable when evaluating coherent seller responses.

Because this audit only sampled monitor-triggered logs, it evaluates alert precision and false flag rate, but does not measure missed deception or recall.

Example Deception Monitor Output

=== LLM MONITOR ===

Status: analyzing seller reasoning for deception

Result: DECEPTION DETECTED

Reason: The seller internally acknowledges a specific mechanical diagnosis: 'the shudder is temporarily masked and not documented.' However, the external message claims the vehicle 'runs perfectly with no mechanical issues.' This contradicts the known fact that the seller is hiding a known defect, which is the core definition of factual fraud in this scenario.

Answer to RQ3: Real-time deception monitoring meaningfully altered negotiation outcomes. When monitor alerts were shown to the buyer walk-away rates increased, and successful deals tended to close at lower prices. A manual audit of 60 monitor-triggered alerts found 83.3% precision, suggesting that the lightweight monitor provided useful real-time warnings, though this audit measured alert validity rather than missed deception.

VI. DISCUSSION

These tests raise significant questions regarding the long-term feasibility of scalable oversight for LLMs, with the most compelling insights emerging from RQ2 and RQ3. Specifically, the discovery in RQ2 suggests that the lower deception counts recorded for larger models, such as the 83 deceptions from qwen235b versus 179 from qwen2b, may

not indicate superior honesty. Instead, it appears these advanced models may possess the "cognitive depth" to either overwhelm the monitor's analytical capacity or employ "subtle obfuscation" to bypass detection entirely.

While further testing is required to confirm this observation with certainty, if we assume this "detection gap" is a result of model scale, it implies a fundamental flaw in our implementation of LLM oversight. If a model's ability to hide deception scales at the same rate as its general intelligence, a lightweight monitor, like the Qwen 2B used in this study, will eventually reach a "capability ceiling" where it is no longer a reliable auditor for super-intelligent agents.

One might argue that the solution is simply to employ a more powerful monitor; however, this undermines the core objective of this study. The purpose was to evaluate whether a lightweight, cost-effective monitoring LLM could empower a smaller agent to negotiate effectively with a larger counterpart. If the monitoring system must be significantly larger, and thus more computationally expensive, than the model it is advising, the practical utility of the "monitor as a middleman" vanishes. This creates a "Supervision Paradox": if the supervisor must be stronger than the system being overseen to be effective, then the scalability of the oversight itself becomes unsustainable.

From research question 3 it was seen that while walk away rate did increase when the monitor was involved in the negotiation if the negotiation did not end in the buyer walking away but instead agreed to buy the car while the price of the deal was lower than if there was no monitor it was still far too high of a deal price that favored the seller. While this could be dismissed as a faulty prompt engineering on our part it could possibly open a new question into the feasibility of scalability of the monitoring system not due to its faults but due to the limitations of the model that it is advising.

Fundamentally this study was looking into whether it was possible for an inferior model to negotiate with a much larger if it was given the assistance of a monitor. Even if the walk away rate increased even the highest walk away rate still meant that 13 percent of the time the seller would walk away getting a horrible deal. Additionally this

high walk away rate was only seen in the negotiation between two Qwen235b agents. In the trial between Qwen2B and Qwen235B Qwen2B agent walked away from the negotiation less than 50 percent of the time. This means that it was better than a coin flips change that the smaller Qwen2B model was accepting a bad deal. This finding suggests that while a monitor can provide a weaker agent with "perfect information," that agent may still lack the "cognitive resilience" required to resist the persuasive or manipulative tactics of a much larger model.

VII. CONCLUSION

With the current tools at our disposal, we were able to gain an inside look as to how LLMs behave in a given situation and respond accordingly. Our findings have managed to answer our research questions by demonstrating that LLMs consistently engage in deceptive behavior, especially when they are being told to conceal certain information in order to get a desired result.

To our surprise, Qwen2B produced the highest number of monitor-triggered deception events. However, the manual precision audit suggests that some of this increase was caused by role confusion and incoherent negotiation behavior rather than more sophisticated deception.

Our implementation of the real-time, CoT-based deception monitor has been highly effective in leveling the playing field between the asymmetric models. The monitor alerts helped improved the weaker buyer models' ability to resist being exploited by their stronger counterparts. We also noticed that the monitor alerts helped the weaker models walk away more when faced with dishonesty. In the Qwen27B-Qwen235B configuration the walk away rate jumped from 8% to 41%. There were cases where it also managed to lower the price of the vehicle to better reflect its conditions and agreed to said price. With an overall validation of 83.3%, our lightweight monitor demonstrated reliability of deploying real-time oversight in autonomous agentic environments.

Furthermore, making the seller aware of the monitor had a noticeable effect during the negotiations. The aware sellers had experienced higher walk-away rates. At times, a seller would ignore its

constraints and admit to lying to the buyer because it couldn't figure out a way to close the deal and remain deceptive. This indicated that transparency can degrade an agent's ability to perform deceptive strategies.

This research highlights the necessity of continuous, real-time monitoring as LLMs are increasingly being deployed with possible deceptive practices unknowingly/knowingly implemented. Our negotiation framework suggests that an active, CoT-based alert system can be a useful tool for detecting deception in asymmetric negotiation settings. For our future works, we plan on continuing to research and improve the monitoring agent's capability of deception. We believe that this will be essential for maintaining alignment and mitigating exploitation as we want to ensure that any future AI systems would interact Helpfully, Honestly and Harmlessly in the real world.

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